Elsayed Mustafa

AI & Automation Engineer

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Completed military service LinkedIn GitHub Upwork Portfolio

Summary

AI & Automation Engineer with 2+ years of experience shipping production automation and LLM systems end to end: marketplace intelligence platforms, bilingual client CRMs, and a local-first AI copilot with scoped retrieval (RAG) and grounding controls. Works in Python, FastAPI and Supabase/PostgreSQL, and owns each solution from requirements and data modeling through integration, testing and delivery.

Experience

AI & Automation Engineer, *BlueBug* 02/2026 – Present | Remote

- Build and maintain three internal AI and automation products in Python, FastAPI and Supabase.
- Built a review-intelligence system that tracks about 10,000 products and keeps 800,000 reviews reconciled against 1.1M snapshots.
- Developed a rating model that gets the exact star rating right 80.7% of the time on products it has never seen.
- Built a Solana research platform that holds dashboard queries at 0.84 seconds against 10+ GB of data, with every strategy tested on paper first behind two independent safety switches.

AI Automation Developer, *Upwork* 06/2024 – Present | Remote

- Delivered 6+ client platforms end to end, from the first call to launch, and earned Upwork Rising Talent.
- Rebuilt a bilingual field-service CRM in Bahrain: moved 17,000+ records into a secured database and cut the slowest page from 5.6 seconds to 0.56 (10× faster).
- Built inventory and marketplace-pricing platforms across Amazon, eBay and Shopify covering 25,000+ items.
- Built a lead pipeline covering 2,662 towns that resumes exactly where it stopped.

Education

B.Sc. in Computer and Information Science, *Ain Shams University* 2020 – 2024

Scientific Computing Department

- Graduation project, Data-Driven English Practice App: processed datasets to simulate conversation and grammar correction, fine-tuned the model, and analyzed results to improve accuracy and feedback.

Skills

AI & Machine Learning

- LLM, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Ollama, Whisper, LightGBM, scikit-learn, NLP, Model Evaluation

Python & Backend

- Python, FastAPI, REST APIs, Authentication, Webhooks

Automation & Scraping

- Playwright, Selenium, BeautifulSoup, Google Apps Script, Data Pipelines

Databases & Data

- PostgreSQL, Supabase, SQL, SQLite, Pandas, NumPy

APIs & Integrations

- Amazon SP-API, eBay API, Google Workspace APIs, Green API, Telegram Bots, Solana APIs

Business Systems

- CRM Platforms, Inventory & Pricing Systems, Role-Based Access Control, Audit History, Operational Dashboards

Projects

Freelancer Meeting Copilot 07/2026

Personal Project | Python, React/TypeScript, Tauri, SQLite, Ollama, Whisper

- Built a private AI assistant for client calls that runs entirely on the user's machine with local model serving, so no call data ever leaves the laptop.
- Scoped retrieval (RAG) so answers come only from that client's own notes, and nothing is treated as a fact until it is confirmed, which keeps the model from inventing details.

Amazon ASIN & Review Intelligence Platform

BlueBug | Python, Playwright, Supabase, Google Sheets APIs

- Built six Amazon scraping and auditing tools for ASIN details, review collection, variation analysis, listing health, and gap detection across multiple marketplaces.
- Added resumable workflows, captcha/TOTP handling, account rotation, retry logic, and synchronized reporting through Supabase and Google Sheets.

Amazon Rating Prediction & Simulation Tool

BlueBug | LightGBM, Python, Pandas, SQLite, JavaScript

- Developed a 37-feature LightGBM model to estimate Amazon displayed ratings, achieving 80.7% exact-match accuracy and 0.034 MAE on unseen ASINs.
- Built an interactive application for rating estimation, review-impact analysis, target-rating simulations, confidence reporting, and downloadable reports.

Solana Market Intelligence & Paper-Trading Platform

BlueBug | Python, WebSockets, Supabase, Streamlit, Machine Learning

- Built a real-time Solana scanner and paper-trading system integrating DexScreener, Birdeye, GeckoTerminal, Helius WebSockets, and Jupiter quotes.
- Implemented token discovery, liquidity and momentum filters, risk controls, backtesting, ML-assisted entry and exit analysis, experiment tracking, and a Streamlit dashboard.

Bahrain Pest Control CRM & Field Operations Platform

Upwork Client | Supabase, PostgreSQL, Apps Script, Green API

- Built a bilingual field-service CRM and migrated 17,000+ records into 17 secured Supabase tables, reducing key server reads from approximately 5.6 seconds to 0.56 seconds.

Sync Corp Operations & Marketplace Pricing Platform

Upwork Client | Python, Supabase, PostgreSQL, Amazon SP-API, eBay

- Built an inventory and marketplace-pricing platform with multi-format supplier ingestion, GTIN validation, product matching, transaction-safe workflows, and 162 passing automated tests.

Victoria Lead Generation & Email Enrichment Pipeline

Upwork Client | Python, Google Maps, Apify, Web Scraping, Data Enrichment

- Built a resumable scraping and enrichment pipeline covering 2,662 Victoria towns, with lead classification, website and email extraction, deduplication, retry handling, and structured QA reports.

Certificates

Machine Learning Specialization — DeepLearning.AI

Advanced SQL Querying — Maven Analytics

Associate Data Analyst in SQL — DataCamp

Natural Language Processing with Classification and Vector Spaces — DeepLearning.AI

Languages

Arabic
Native

English
Professional Proficiency